



FLIGHT STAND 500

THRUST STAND AND SOFTWARE FOR MOTOR AND PROPELLER CHARACTERIZATION

Key Specs

- Thrust: $\pm 5,000$ N
- Torque: $\pm 1,500$ Nm
- RPM: up to 16,800 RPM
- Voltage: up to 1,000 V
- Current: up to 500 A

See next page for full product specifications.

Applications

- Propulsion system characterization and comparison
- Endurance testing
- Reliability testing
- Motor heat testing
- Maintenance and quality control
- Flight replay
- Real-time dynamic testing
- Aerodynamic testing

Description

The Flight Stand 500 thrust stand precisely characterizes the performance of UAV motors and propellers. Included is the hardware, electronics and software required to measure your propulsion system's thrust, torque, RPM, current, voltage, temperature, electrical and mechanical power, propeller efficiency, motor efficiency, overall efficiency, and more.



Note: motor and propeller not included

Figure 1: Flight Stand 500

PRODUCT SPECIFICATIONS

	Specification	Value
Basic Information	Operating temp & humidity	0 to 40° C, <70% RH suggested
	Storage temp & humidity	23° C, 20% RH suggested
	Dimensions	104" (H) x 27" (W) x 39" (D) 2.66 m (H) x 0.68 m (W) x 1 m (D)
	Net weight	400 kg
	Input power / Output power	Two 9 V, 2 A wall adaptors for DAQ One 12 V, 2 A wall adapter for Fiber Optic RPM sensor System must be properly grounded
Measurement Information	Sampling rate	1,000 Hz
	Wireless testing	Yes
	Calibration methodology	Tyto Methodology (V2.0)
	Thrust matrix generation	ASTM E74-derived protocol; >200 points push & pull
	Torque matrix generation	ASTM E2624-derived protocol; >200 points CW & CCW
Thrust	Range	±5,000 N
	Resolution	5 N
	Accuracy	0-500 N: ±3.75 N 500-5,000 N: 0.75% of the measured value
Torque	Range	±1,500 Nm
	Resolution	1.5 Nm
	Accuracy	0-150 Nm: ±1.5 Nm 150-1,500 Nm: 1% of the measured value
Voltage and Current	Voltage range	25 to 1,000 V
	Voltage resolution	0.001 V
	Voltage accuracy	0.5% from 25 to 1,000 V
	Current range	5 to 500 A
	Current resolution	0.001 A
	Current accuracy	±1% of measured value from 50 to 500 A
Temperature (Thermocouple)	Range	3 to 200° C
	Resolution	1 K
	Accuracy	±1.1 K

PRODUCT SPECIFICATIONS

	Specifications	Value
Fiber Optic RPM Sensor	Maximum RPM	Two-blade propeller: 16,800 RPM Three-blade propeller: 11,200 RPM Four-blade propeller: 8,400 RPM
	Resolution	1 RPM
	Operating environment	Indoor and outdoor
	Operational distance	20 to 100 mm
Airspeed Pressure Sensor	Range (differential pressure)	10 to 498 Pa
	Accuracy (differential pressure)	±2.25%
	Range (airspeed)	2 to 28 m/s
	Resolution (airspeed)	0.1 m/s
	Pitot tube	Not included
	Max distance between pitot tube and sensor	1 m
	Humidity	up to 95% RH - non-condensing
DAQ	Range	0.02 to 5 V single ended input, 100 mA max
	Resolution	0.001 V, 24 bit ADC resolution, 16 bit ENOB
	Accuracy	±0.4% F.S. (±0.02 V)
	Analog pin out	3.5 mm TRRS, 100 mA max
	# of available analog ports for additional sensors	4
	Pulse signal	Rising edge, TTL compatible
	Pulse pin out	3.5 mm TRRS, 5 V, 30 mA max
	# of available pulse input ports for additional sensors	3
Control	Number of outputs	2 PWM for ESC, 6 PWM for servo
	Pin out	3.5 mm TRRS, 5 V, 30 mA max
	PWM output protocol / frequency	Standard PWM 50, 100, 200, 300, 400, 500 Hz DSHOT 150, 300, 600 OneShot 42, 125 MultiShot
	Other Control	CAN-Bus through API
System Requirements	Windows Version	Windows 10: Version 2004 or newer, Windows 11
	Storage	NVMe or PCIe SSD Required
	CPU	8 cores+



CALIBRATION AND VERIFICATION

Tyto Methodology for Calibration Verification (V2.0)

Overview

Our matrix generation process is inspired by elements of ASTM E74 and ASTM E2624 and adapted for the thrust and torque measurements used in UAV propulsion testing.

The process uses Tyto-developed metrology equipment to apply controlled thrusts and torques representative of typical UAV test conditions. This includes extensive loading scenarios covering the whole measurement range with over 20 points for the Standard version and over 200 points for the Pro version.

To achieve this, custom machines are used to apply precise forces automatically in both thrust and torque. The loading sequences generate a matrix for the measurement system, which accurately transforms the multiple strain gauge outputs to temperature-compensated thrust and torque measurements.

Calibration is then performed using Tyto's internal methodology, which relies on dead weights and precision load cells across representative portions of the tool's measurement range. This process is built around controlled loading, repeatability, and traceability. All force, voltage, and current sensors use traceable equipment that is itself recalibrated by external accredited labs at regular intervals.

This verification data is available as a PDF upon request. Contact our sales team for more information.

Technical definitions:

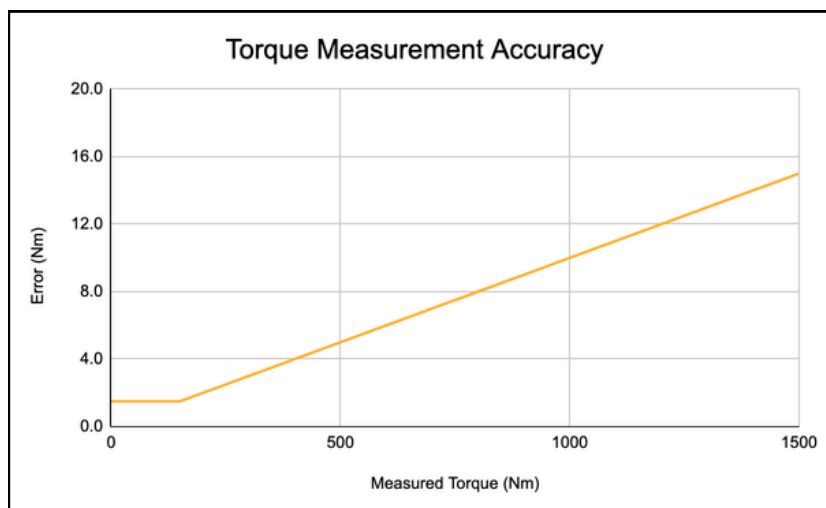
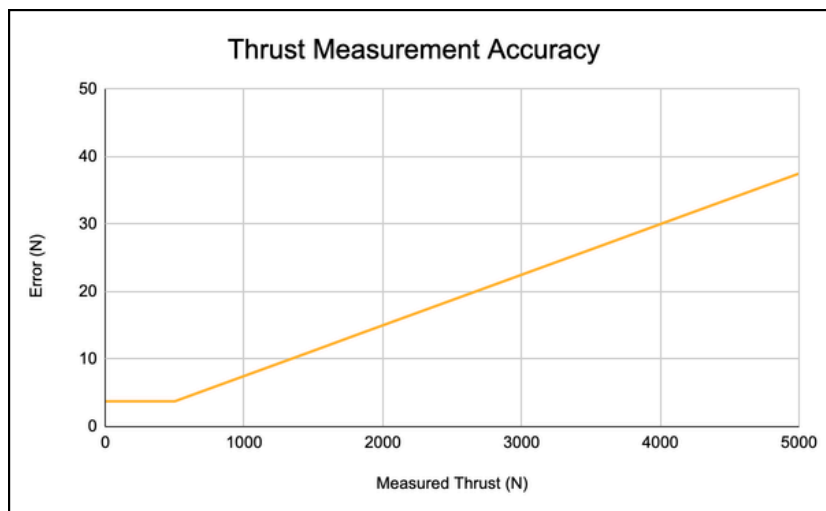
Accuracy: The maximum allowed average error observed during calibration. The calibration is performed using dead weights and precision load cells over a range representative of the tool's intended measurement range. The calibration is done with a single force, after the matrix generation process.

Resolution: Three times the standard deviation of the noise floor measured during a 1-second static recording at the full sample rate. Improved effective resolution may be achieved when exporting filtered data.

ACCURACY

Measurement accuracy was our top priority when designing the Flight Stand. Careful load cell design and calibration has allowed us to offer traceable, industry-leading accuracy with minimal measurement error.

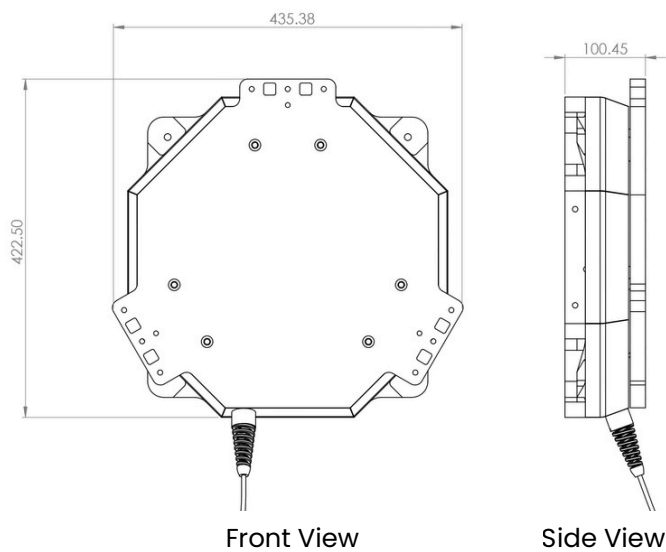
Our Standard thrust stands have a constant error across the measurement range, while our Pro stands have an error proportional to the measured value, offering even higher accuracy.



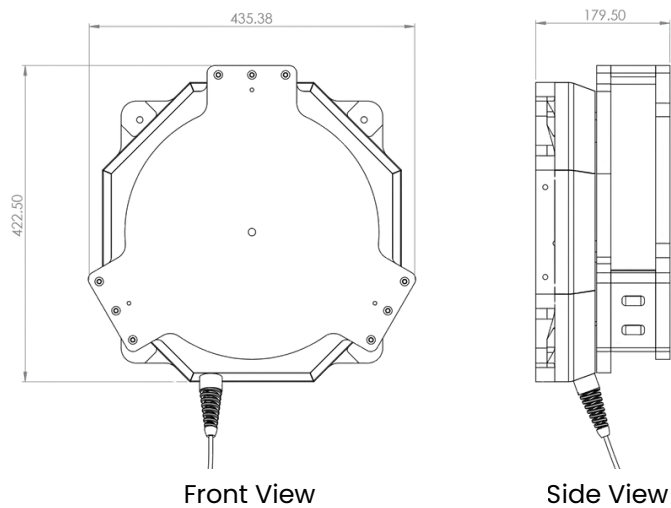
TECHNICAL DRAWINGS

Force Measurement Unit:

Without Motor Mounting Plate



With Motor Mounting Plate



Stand Structure:

